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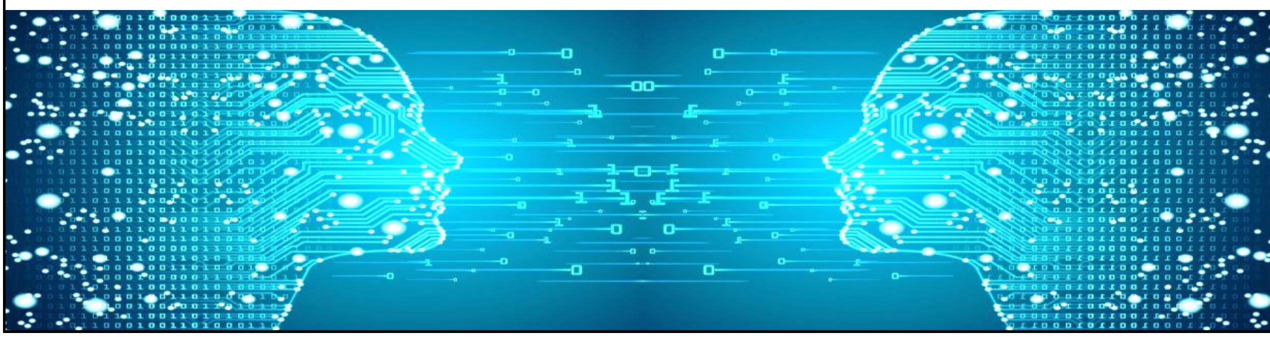
Machine Learning



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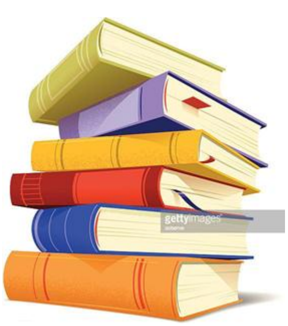
Chapter 7

Linear Discriminant Analysis



- Linear Discriminant Analysis (LDA)
- Example

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Linear Discriminant Analysis

What It Is

- **Origins:** Discussed by statistical pioneer **Ronald Fisher in 1936**.
- **Core Goal:** To find a **linear combination of features** that maximizes the separation (discrimination) between two or more classes.
- **Nature:** LDA is a **generalization of the Fisher linear discriminant**.
- **Output:** It produces exactly as many **linear functions** as there are classes.
- **Classification:** The predicted class for a new instance is the one that yields the **highest value** for its linear function.

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Linear Discriminant Analysis

Key Benefits

- **Classification:** Separates data instances into distinct classes.
- **Dimensionality Reduction:** The resulting linear combination can be used to reduce data dimensionality.
  - The purpose of dimensionality reduction is to eliminate features of little value (noise) and retain the most important information.

Practical Applications

- **Finance:** Used to explain the **bankruptcy or survival of a firm**.
- **Pattern Recognition:** Often used in **face recognition problems** to reduce dimensions.

Prediction

- The **linear discriminant function** used in **LDA**:

$$g(x) = w^T x + w_0$$

  - w: weight vector (computed as $\Sigma^{-1}(\mu_1 - \mu_2)$)
  - w_0: bias term (intercept)
  - x: input feature vector

- The **decision rule** of LDA:

$$\text{Predicted class} = \begin{cases} 0, & if\ g(x) > 0 \\ 1, & if\ g(x) \le 0 \end{cases}$$

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Linear Discriminant Analysis

Steps to Determine LDA Weights

1. **Calculate Class Means (μ_1, μ_2):** Compute the mean vector for each class.
2. **Calculate Individual Covariance Matrices:** Compute the covariance matrix for each class.
3. **Calculate Pooled Covariance Matrix (Σ)**
4. **Calculate Prior Probabilities:** Determine the proportion of records belonging to each class.
5. **Calculate Weights ($W = \Sigma^{-1}(\mu_1 - \mu_2)$)**

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Example

Predicting Smartphone Preferences

- **Scenario:** Predict the type of smartphone a customer is interested in.
- **Classes (C):** Two classes defined:
  - C_1: "Apple" (represented as 0)
  - C_2: "Samsung" (represented as 1)
- **Features (Numeric Variables):**
  - x_1: Age of the customer
  - x_2: Income of the customer
- **Goal:** Understand how LDA calculates the weights necessary to classify new customers based on their age and income.

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Example

Final Discriminant Weights (W)

Weight Component	Value for Class 0 (Apple)	Value for Class 1 (Samsung)
W_0 (Intercept/Bias)	-71.52	-59.38
W_1 (Age coefficient)	1.406	1.823
W_2 (Income coefficient)	0.0101	0.0064

- **Interpretation:** These weights define two linear discriminant functions (one for each class). For a new customer with Age (x_1) and Income (x_2), we calculate:

$$Score_{Apple} = W_{0,Apple} + W_{1,Apple} \cdot x_1 + W_{2,Apple} \cdot x_2$$

$$Score_{Samsung} = W_{0,Samsung} + W_{1,Samsung} \cdot x_1 + W_{2,Samsung} \cdot x_2$$

- The instance is assigned to the class with the **highest score**.

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Summary

Concept	Key Takeaway
Definition	LDA is a supervised technique aiming to find a linear combination of features that maximizes the separation between classes.
Goals/Applications	Primary uses are Classification (predicting the highest scoring linear function) and Dimensionality Reduction (reducing data features while retaining critical information).
LDA Output	Produces one linear function for each class (e.g., Apple vs. Samsung). The new instance is assigned to the class with the highest function value.
Mathematical Core	Calculating the discriminant weights (W) requires computing the Mean Vectors (μ_i) and the Pooled Covariance Matrix (which estimates a common covariance across classes).
Example Interpretation	The final weights (W) define the discriminant functions based on features (like Age x_1 and Income x_2). The classification decision is purely linear.

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